

a carrier frequency of 434 MHz and transmits the data through the antenna. The concept of the ASK signal is shown in Fig. 3.

RF reception. The ASK receiver module (Fig. 4) receives the 434MHz signal

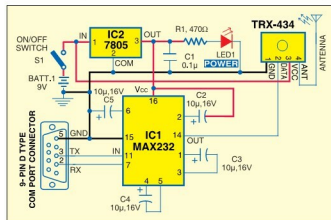


Fig. 2: Circuit of transmitter section

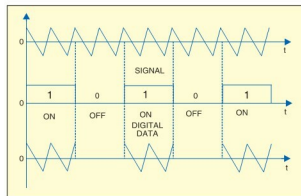


Fig. 3: ASK-modulated signal

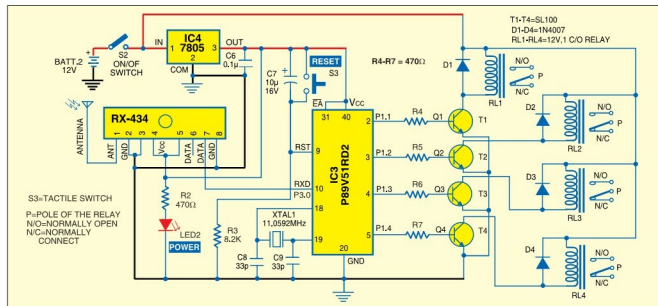


Fig. 4: Circuit of receiver section

TABLE II
ASK Module Specifications

Details	Specifications
Frequency	434 MHz
Transmitter	Saw-filter-based ASK hybrid transmitter
Transmitter supply voltage	3-12V
Receiver	ASK super heterodyne receiver with PLL synthesiser and crystal oscillator
Receiver supply voltage	5V
Receiver supply current	2.3 mA
Output power	4-16 dBm
Data rate	200 bps to 3 kbps depending on the supply

and demodulates it. The baseband signal is a 10-bit digital signal with the first bit as start bit, the following

eight bits as data bits and the last bit as stop bit. The received data is sent to the microcontroller. The RXD pin (p3.0) of P89V51RD2 microcontroller is connected to the data pin of the 434MHz ASK module receiver. So the P89V51RD2 detects the transmitted data at its pin 10 (RXD).

Decoding of the received signal with P89V51RD2. The P89V51RD2 is an 80C51 microcontroller with 64kB flash, 1024 bytes of data RAM, 32 input/output (I/O) ports, three 16-bit timers/counters and two pins for serial data transmission and reception. Timer 1 is used for serial communication. It is operated with an 11.0592MHz crystal.

A key feature of the P89V51RD2 is its X2 mode option. You can choose to run the application with the conventional 80C51 clock rate (12 clocks per machine cycle) or select the X2 mode (six clocks per machine cycle) to achieve twice the throughput at the same clock frequency.

The Flash program memory supports both parallel programming and